

Python在财务中的综合应用

行业财务报表采集

1 案例背景

2 任务要求

3 任务解析

中捷资源投资股份有限公司近年一直处于亏损的状态，为了了解行业其他企业的经营情况，需要财务人员关注同行业的年度财务报表数据做分析，财务任务可以通过编写行业报表采集的python爬虫采集程序来实现同行业全部企业的报表采集。

1 案例背景

2 任务要求

3 任务解析

编写Python代码，在新浪财经采集纺织机械行业下全部企业近10年的资产负债表、利润表和现金流量表的年报数据。

纺织机械行业网址为：https://vip.stock.finance.sina.com.cn/mkt/#new_fzjx

以标准股份为例，三大报表网址为：

https://vip.stock.finance.sina.com.cn/corp/go.php/vFD_FinanceSummary/stockid/600302/displaytype/4.phtml?source=fzb

1 案例背景

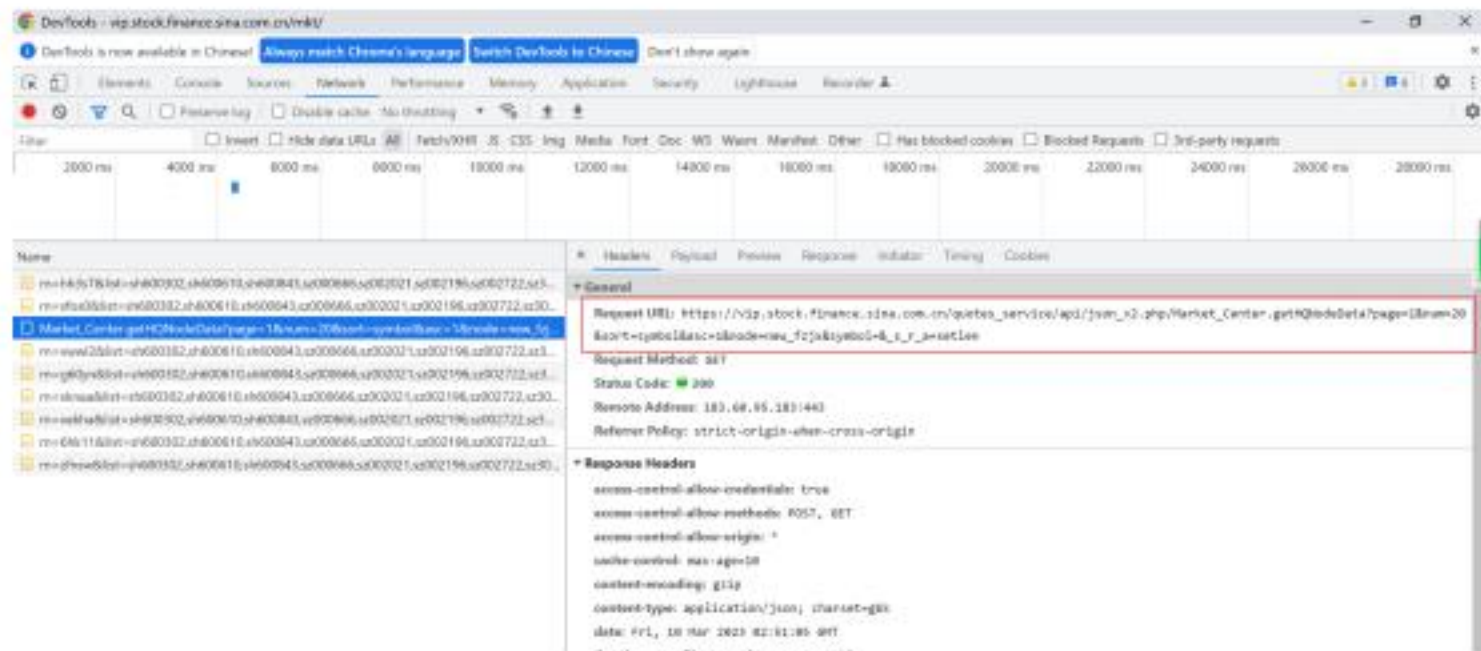
2 任务要求

3 任务解析

任务编码主思路：

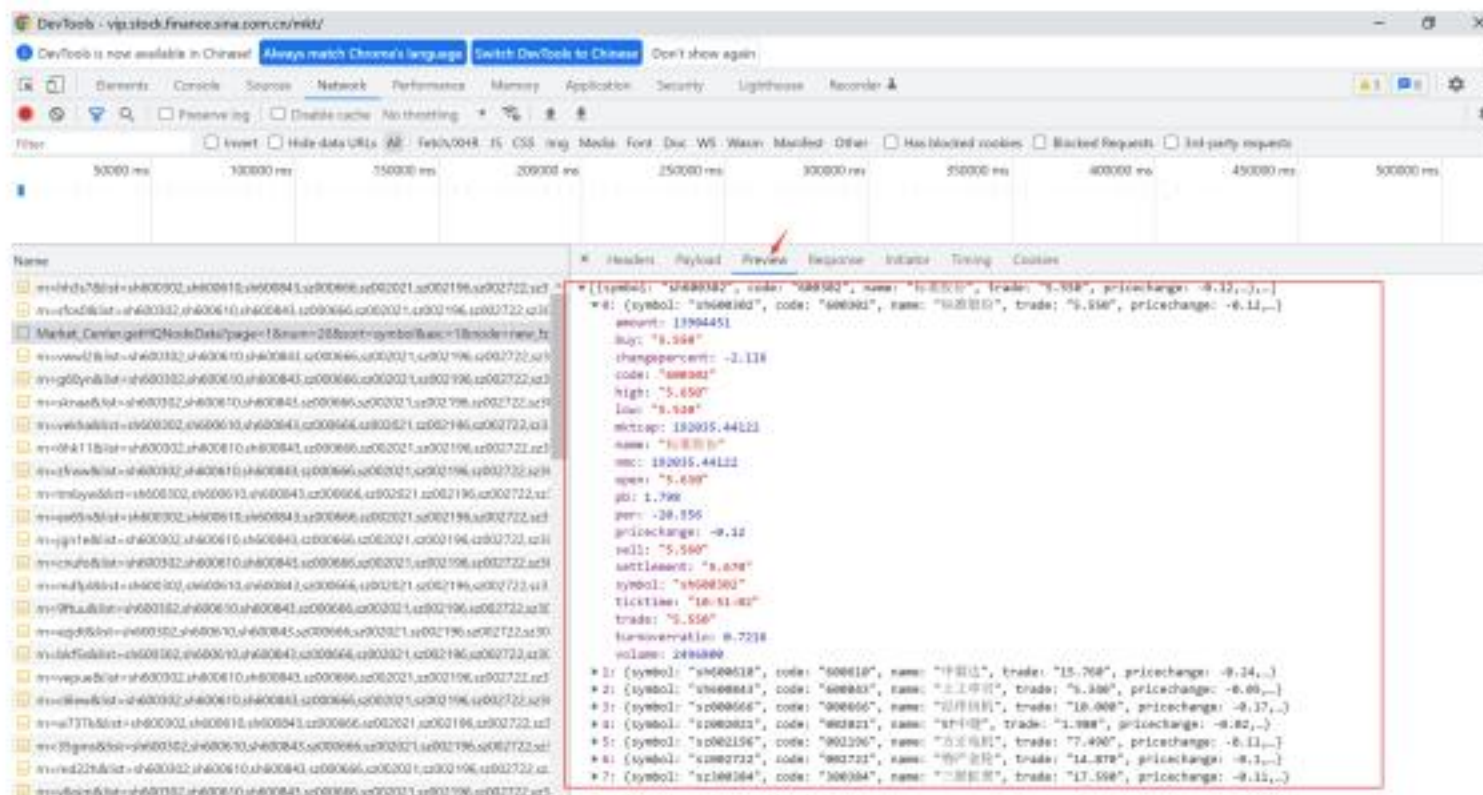
1. 打开纺织机械行业页面，在开发者工具下，找到该行业下企业的查询接口对应的url，具体url信息如下：

`https://vip.stock.finance.sina.com.cn/quotes_service/api/json_v2.php/Market_Center.getH
QNodeData?page=1&num=20&sort=symbol&asc=1&node=new_fzjx&symbol=&_s_r_a=s
etlen`



任务编码主思路:

2. 在开发者工具下，切换到Preview页签，可看到请求返回的数据是json格式的，通过对数据分析可知返回的数据是纺织机械行业下的全部公司，包含公司股票代码、公司代码、公司名称等信息：



任务编码主思路：

3. 以标准股份为例，打开三大报表数据的网页，打开开发者工具，切换到年报数据。

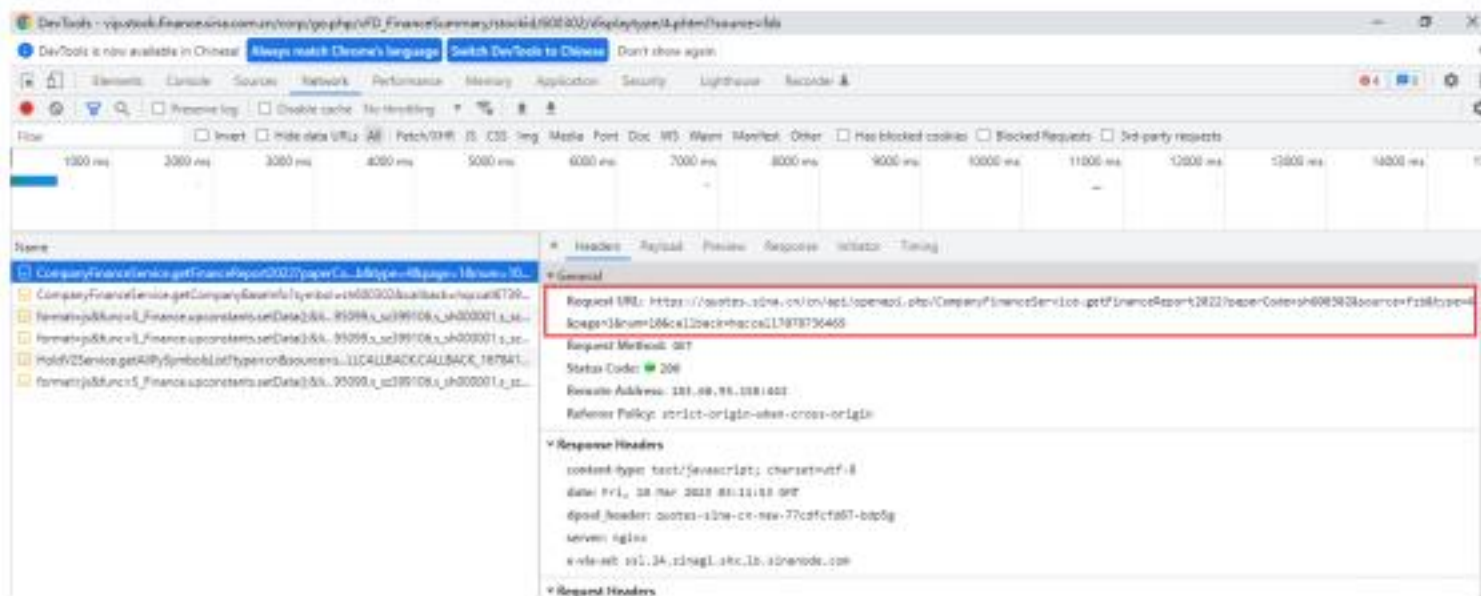
The screenshot displays the Sina Finance website's financial summary for Standard Shares (标准股份). The page shows various financial metrics, including the stock price (5.56), market capitalization (18.24 billion), and key financial indicators. The Chrome DevTools Network tab is open, showing the network activity for the page. The 'Network' tab is selected, and the 'Filter' bar is set to 'All'. The network log shows several requests, including 'CompanyFinanceService.getCompanyDetailBySymbol' and 'FormatJsonService.FinanceUpconcreteData'. The 'Timeline' view at the bottom shows the sequence of requests and their durations.

Name	Status	Type	Initiator	Size	Time	Waterfall
CompanyFinanceService.getCompanyDetailBySymbol	200	script	formatJsonService.FinanceUpconcreteData	152 B	152 ms	
FormatJsonService.FinanceUpconcreteData	200	script	formatJsonService.FinanceUpconcreteData	152 B	152 ms	

任务编码主思路：

4. 找到标准股份公司的资产负债表年报的查询接口的url，具体url信息如下：

<https://quotes.sina.cn/cn/api/openapi.php/CompanyFinanceService.getFinanceReport2022?paperCode=sh600302&source=fzb&type=4&page=1&num=10>



任务编码主思路：

5. 用同样的方法找到标准股份公司的利润表年报的查询接口的url，具体url信息如下：

<https://quotes.sina.cn/cn/api/openapi.php/CompanyFinanceService.getFinanceReport2022?paperCode=sh600302&source=lrb&type=4&page=1&num=10>

6. 用同样的方法找到标准股份公司的现金流量表年报的查询接口的url，具体url信息如下：

<https://quotes.sina.cn/cn/api/openapi.php/CompanyFinanceService.getFinanceReport2022?paperCode=sh600302&source=llb&type=4&page=1&num=10>

7. 用同样的方法找到其他公司的三大报表年报的查询接口的url，分析后发现，接口url一样，区别在于不同企业再url中传入的paperCode是不同的企业股票代码。

任务编码主思路：

8. 根据以上获取的url，调用requests库的get方法来发送请求，通过text获取网页相应内容，并将获取的内容转换为json格式存储。

```
* Headers Payload Preview Response Initiator Timing
▼ /*<script>location.href="//sina.com";</script>*/ hqccall4055862589({result: {status: {code: 0}, data: {report_count: "25",...}}
▼ result: {status: {code: 0}, data: {report_count: "25",...}}
▼ data: {report_count: "25",...}
  report_count: "25"
  report_date: [{date_value: "20211231", date_description: "2021年报", date_type: 4},...]
  report_list: {...}
    ▼ 20121231: {rType: "合并期末", rCurrency: "CNY", data_source: "定期报告", is_audit: "是", publish_date: "20140303",...}
      data: [{item_field: "", item_title: "流动资产", item_value: "", item_display_type: 1, item_display: "大类",...}]
        ▼ [0 - 99]
          ▶ 0: {item_field: "", item_title: "流动资产", item_value: "", item_display_type: 1, item_display: "大类",...}
          ▶ 1: {item_field: "CURFDS", item_title: "货币资金", item_value: "244537261.290000", item_display_type: 2,...}
          ▶ 2: {item_field: "SETTRESDEPO", item_title: "结算备付金", item_value: null, item_display_type: 2,...}
          ▶ 3: {item_field: "PLAC", item_title: "拆出资金", item_value: null, item_display_type: 2, item_display: "小类",...}
          ▶ 4: {item_field: "TRADFINASSET", item_title: "交易性金融资产", item_value: null, item_display_type: 2,...}
          ▶ 5: {item_field: "PURCFRESAAASSET", item_title: "买入返售金融资产", item_value: null, item_display_type: 2,...}
          ▶ 6: {item_field: "DERIFINAAASSET", item_title: "衍生金融资产", item_value: null, item_display_type: 2,...}
          ▶ 7: {item_field: "NOTESACCORECE", item_title: "应收票据及应收账款", item_value: "228820190.860000",...}
          ▶ 8: {item_field: "NOTESRECE", item_title: "应收票据", item_value: "33835235.390000", item_display_type: 3,...}
          ▶ 9: {item_field: "ACCORECE", item_title: "应收账款", item_value: "194984955.470000", item_display_type: 3,...}
          ▶ 10: {item_field: "RECFINANC", item_title: "应收款项融资", item_value: null, item_display_type: 2,...}
          ▶ 11: {item_field: "PREP", item_title: "预付款项", item_value: "4741933.210000", item_display_type: 2,...}
          ▶ 12: {item_field: "DIVIDRECE", item_title: "应收股利", item_value: null, item_display_type: 2,...}
          ▶ 13: {item_field: "INTERECE", item_title: "应收利息", item_value: null, item_display_type: 2,...}
          ▶ 14: {item_field: "PREMRECE", item_title: "应收保费", item_value: null, item_display_type: 2,...}
          ▶ 15: {item_field: "REINRECE", item_title: "应收分保账款", item_value: null, item_display_type: 2,...}
          ▶ 16: {item_field: "REINCONTRESE", item_title: "应收分保合同准备金", item_value: null, item_display_type: 2,...}
          ▶ 17: {item_field: "EXPOTAXREBARECE", item_title: "应收出口退税", item_value: null, item_display_type: 2,...}
          ▶ 18: {item_field: "SUBSRECE", item_title: "应收补贴款", item_value: null, item_display_type: 2,...}
          ▶ 19: {item_field: "MARGRECE", item_title: "应收保证金", item_value: null, item_display_type: 2,...}
          ▶ 20: {item_field: "INTELRECE", item_title: "内部应收款", item_value: null, item_display_type: 2,...}
          ▶ 21: {item_field: "INTELRECE", item_title: "内部应收款", item_value: null, item_display_type: 2,...}
          ▶ 22: {item_field: "OTHERRECE", item_title: "其他应收款", item_value: "9215549.220000", item_display_type: 2,...}
          ▶ 23: {item_field: "OTHERRECE TOT", item_title: "其他应收款(合计)", item_value: "9215549.220000",...}
```

任务编码主思路：

9. 通过解析json数据，按数组存放报表数据。

10. 通过pandas的to_excel，将采集到的报表保存到excel中。

任务实现代码：

参考教材中的代码答案



感謝
ขอบคุณ
terima kasih
ありがとう
Thanks
谢谢